

Test Plan (fixes for MC open cases)

1. Case 00277212 : Viper Map Race Condition

Background:

When Mastercard ran ~3000 CLI executions in parallel, they noticed that the CxOne CLI would intermittently terminate with a fatal message “concurrent map read and map write”.

RCA

The AST CLI uses a Go configuration management library called Viper. Viper’s internal map that stores configuration values is not safe for concurrent read and write operations across goroutines without explicit synchronization. In the CLI, there is a goroutine that attempts to read a configuration string immediately after startup, and sometimes, that read can happen while Viper is still writing to it’s internal map. The Go runtime detects this and kills the CLI with the fatal message.

Objective: Verify that the CLI is able to function without race condition crashes in similar conditions (heavy CPU usage with parallel CLI executions)

TC-1.1 Regression: normal startup for single runs. Run cx scan create and verify the fix did not break normal operation.

Expected: Scan completes normally.

TC-1.2 Parallel executions. Run 10+ scan create commands simultaneously against the same or different projects.

Expected: No process terminates with fatal error: concurrent map read and map write.

Unless we script this like MC, I am not certain we can objectively test this with the right conditions.

2. Case 00277470: Edge case failure in CLI when using --application-name

Background

Mastercard noticed that there were scan failures when they ran the CLI with the --application-name “<application_name>” argument for a project that was *already* associated with the provided application.

RCA

CxOne enforces the following rule around Project <-> Application associations:

“A project cannot be associated with more than one application if any of those applications has a query override.”

However, the CLI failures were an edge case where it was attempting to update the Project <-> Application associations even if they were previously associated. Underneath the covers, it was adding a parameter to a REST call when it shouldn't have.

Objective: Validate that a scan create succeeds under the following conditions:

- The project is already associated with the requested application
- The application has application-level query overrides
- The project is being associated with the same application using the --application-name argument

Pre-condition: A project exists and is already associated with an application that has a query override configured at the application level.

TC-2.1 Project that is already associated with an application (either directly or rules-based).

Run scan create with the --application-name argument. (add --debug for verbose logs)

Expected: Log contains “Application [application_name] : Project [project_name] already associated. Skipping explicit re-association...” and no REST API update (project) call is made.

This can be confirmed by inspecting HTTP calls in the log.

TC-2.2 Repeat-safe. Run the same scan create --application-name command twice in a row for an already-associated project.

Expected: Scan does not fail with “Cannot associate project with multiple applications when one of the applications has query override configured”.

TC-2.3 Regression. Verify scan create works as expected for:

- New projects
- Existing projects
- With application associations
- Without application associations